

HDOC DAY-33

C Programming - Associative Property of Matrices

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Batch	15
Activity	HDOC Day-33

Aim

To write a menu-driven C program using separate functions to verify the associative property of matrix addition and matrix multiplication using variable-length 2D integer arrays.

Problem Statement

1. Read three variable-length 2D integer arrays A, B and C of dimensions $m \times n$ and verify: $(A + B) + C = A + (B + C)$.
2. Read A ($m \times n$), B ($n \times p$), and C ($p \times q$) and verify: $(A \times B) \times C = A \times (B \times C)$.
3. Exit. For each operation, display the matrices obtained on both sides and state whether the associative property is verified.

Algorithm

Main Menu

- Display the three menu options.
- Read the user choice.
- For choice 1, call the matrix-addition associative-property function.
- For choice 2, call the matrix-multiplication associative-property function.
- For choice 3, terminate the program.
- For any other choice, display an invalid-choice message.

Associative Property of Addition

- Read m and n.
- Read matrices A, B and C, each of size $m \times n$.
- Compute $AB = A + B$, then $Left = AB + C$.
- Compute $BC = B + C$, then $Right = A + BC$.
- Display Left and Right.
- Compare corresponding elements. If all are equal, the associative property is verified.

Associative Property of Multiplication

- Read m, n, p and q.
- Read A ($m \times n$), B ($n \times p$), and C ($p \times q$).
- Compute $AB = A \times B$, then $Left = AB \times C$.
- Compute $BC = B \times C$, then $Right = A \times BC$.
- Display Left and Right, both of size $m \times q$.
- Compare corresponding elements. If all are equal, the associative property is verified.

Test Case Table

Test	Input / Choice	Expected Result
1	Choice 1; $m=2, n=2$; valid A, B, C	Both sides are identical; addition is associative.
2	Choice 2; $m=n=p=q=2$; valid A, B, C	Both products are identical; multiplication is associative.
3	Choice 3	Program exits normally.
4	Choice 5	Invalid choice message is displayed.

C Program

```
#include <stdio.h>

void readMatrix(int r, int c, int a[r][c], char name) {
    printf("Enter elements of matrix %c (%d x %d):\n", name, r, c);
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
}

void printMatrix(int r, int c, int a[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) printf("%d\t", a[i][j]);
        printf("\n");
    }
}

void addMatrix(int r, int c, int a[r][c], int b[r][c], int result[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            result[i][j] = a[i][j] + b[i][j];
}

int equalMatrix(int r, int c, int a[r][c], int b[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            if (a[i][j] != b[i][j]) return 0;
    return 1;
}

void multiplyMatrix(int r1, int c1, int c2,
                   int a[r1][c1], int b[c1][c2], int result[r1][c2]) {
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++) {
            result[i][j] = 0;
            for (int k = 0; k < c1; k++)
                result[i][j] += a[i][k] * b[k][j];
        }
    }
}

void associativeAddition(void) {
    int m, n;
    printf("Enter rows and columns (m n): ");
    scanf("%d %d", &m, &n);

    int A[m][n], B[m][n], C[m][n];
    int AB[m][n], BC[m][n], left[m][n], right[m][n];

    readMatrix(m, n, A, 'A');
    readMatrix(m, n, B, 'B');
    readMatrix(m, n, C, 'C');

    addMatrix(m, n, A, B, AB);
    addMatrix(m, n, AB, C, left);
    addMatrix(m, n, B, C, BC);
    addMatrix(m, n, A, BC, right);

    printf("\n(A + B) + C:\n");
    printMatrix(m, n, left);
    printf("\nA + (B + C):\n");
    printMatrix(m, n, right);

    if (equalMatrix(m, n, left, right))
        printf("\nAssociative property of matrix addition is VERIFIED.\n");
    else
        printf("\nAssociative property of matrix addition is NOT verified.\n");
}

void associativeMultiplication(void) {
    int m, n, p, q;
    printf("Enter m n p q: ");
    scanf("%d %d %d %d", &m, &n, &p, &q);

    int A[m][n], B[n][p], C[p][q];
    int AB[m][p], BC[n][q];
    int left[m][q], right[m][q];

    readMatrix(m, n, A, 'A');
```

```

readMatrix(n, p, B, 'B');
readMatrix(p, q, C, 'C');

multiplyMatrix(m, n, p, A, B, AB);
multiplyMatrix(m, p, q, AB, C, left);
multiplyMatrix(n, p, q, B, C, BC);
multiplyMatrix(m, n, q, A, BC, right);

printf("\n(A x B) x C:\n");
printMatrix(m, q, left);
printf("\nA x (B x C):\n");
printMatrix(m, q, right);

if (equalMatrix(m, q, left, right))
    printf("\nAssociative property of matrix multiplication is VERIFIED.\n");
else
    printf("\nAssociative property of matrix multiplication is NOT verified.\n");
}

int main(void) {
    int choice;
    do {
        printf("\n--- HDOC Day-33 Menu ---\n");
        printf("1. Verify associative property of matrix addition\n");
        printf("2. Verify associative property of matrix multiplication\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: associativeAddition(); break;
            case 2: associativeMultiplication(); break;
            case 3: printf("Exiting program.\n"); break;
            default: printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 3);
    return 0;
}

```

Sample Execution

```
--- HDOC Day-33 Menu ---
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 1
Enter rows and columns (m n): 2 2
Enter elements of matrix A (2 x 2):
1 2 3 4
Enter elements of matrix B (2 x 2):
5 6 7 8
Enter elements of matrix C (2 x 2):
1 1 1 1
```

```
(A + B) + C:
7      9
11     13
```

```
A + (B + C):
7      9
11     13
```

Associative property of matrix addition is VERIFIED.

```
Enter your choice: 2
Enter m n p q: 2 2 2 2
A = 1 2 / 3 4
B = 5 6 / 7 8
C = 1 0 / 0 1
```

```
(A x B) x C:
19     22
43     50
```

```
A x (B x C):
19     22
43     50
```

Associative property of matrix multiplication is VERIFIED.

Result

The program successfully reads variable-length 2D arrays, computes both sides of the associative equations, displays the resulting matrices, and verifies the associative properties of matrix addition and matrix multiplication.

Execution Output - HDOC Day-33

The following terminal screenshot shows successful compilation and execution of Option 1, verifying the associative property of matrix addition.

```

    return 0;
}
zsh: parse error near `\'
dhruvpandey@DHRUVs-MacBook-Air ~ %
dhruvpandey@DHRUVs-MacBook-Air ~ %
dhruvpandey@DHRUVs-MacBook-Air ~ %
dhruvpandey@DHRUVs-MacBook-Air ~ %
dhruvpandey@DHRUVs-MacBook-Air ~ % o
zsh: command not found: o
dhruvpandey@DHRUVs-MacBook-Air ~ %
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day33.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clear

dhruvpandey@DHRUVs-MacBook-Air ~ % clang day33.c -o day33
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day33

===== HDOC DAY-33 =====
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit

Enter your choice: 1

Enter rows and columns (m n): 2
3

Enter elements of Matrix A:
2
343
3
4
3
4

Enter elements of Matrix B:
2
354
55
655
3
55

Enter elements of Matrix C:
666
776
44
33
5
6

(A + B) + C =
670    1473    102
692     11     65

A + (B + C) =
670    1473    102
692     11     65

Associative property of matrix addition is VERIFIED.

```